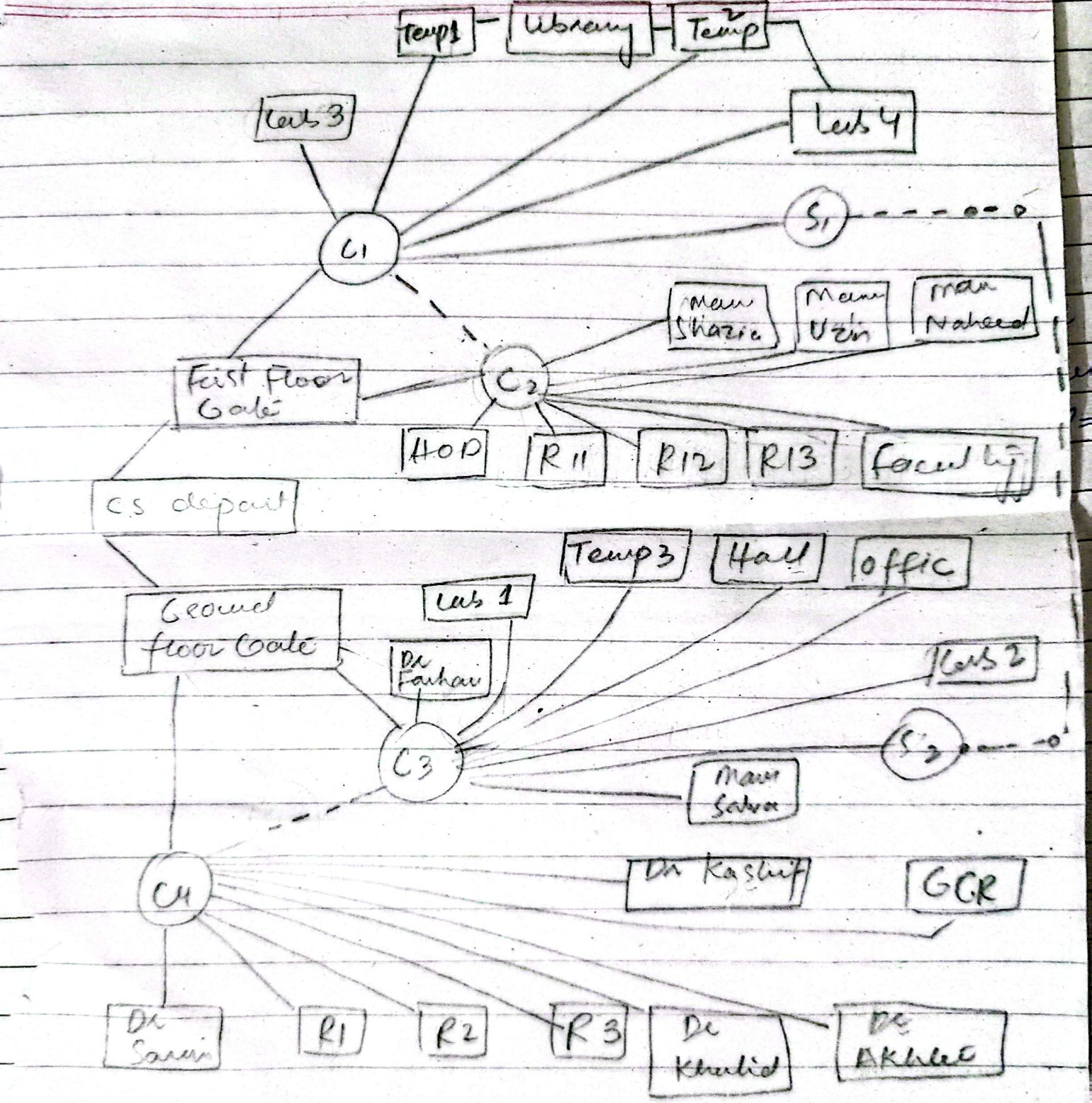
Cs Department Map



Translation of graph (cs depart).

- cs depart → Fiist floor, Second floor
- Fiist Floor → cs depart, C_1, C_2
- C_1 → Fiist floor, Lab 3, Temp 1, Temp 2, Lab 4, C_2, S_1
- Temp 1 → Lobby, C_1
- Lobby → Temp 1, Temp 2
- Temp 2 → C_1, Lobby, Lab 4
- C_2 → Fiist Floor, C_1, HOD, R11, R12, R13, Faculty, Mam Shazia, Mam Uzma, Mam Nabeel.
- S_1 → C_1, S_2
- Ground Floor → cs depart, C_3, C_4
- C_3 → Ground Floor, Dr Farhan, Lab 1, Temp 3, Hall, office, Lab 2, Mam Sadia, S_2, C_4
- S_2 → S_1, C_3
- C_4 → Ground Floor, C_3, Dr Sami, R1, R2, R3, Dr Khalid, Dr Akhter, Dr Kashif, GCR
- Lab 3 → C_1
- Lab 4 → C_1, Temp 4
- Mam Shazia → C_2
- Mam Uzma → C_2
- Mam Nabeel → C_2
- HOD → C_2
- R11 → C_2
- R_{12} → C_2
- R_{13} → C_2

- Dr Farhan → C_3
- Lab 1 → C_3
- Temp 3 → C_3
- Hall → C_3
- office → C_3
- Lab 2 → C_3
- Main Salwa → C_3
- Dr Kashif → C_4
- GCR → C_4
- Dr Sami → C_4
- R_1 → C_4
- R_2 → C_4
- R_3 → C_4
- Dr Khalid → C_4
- Dr Akhlet → C_4
- Faculty → C_2

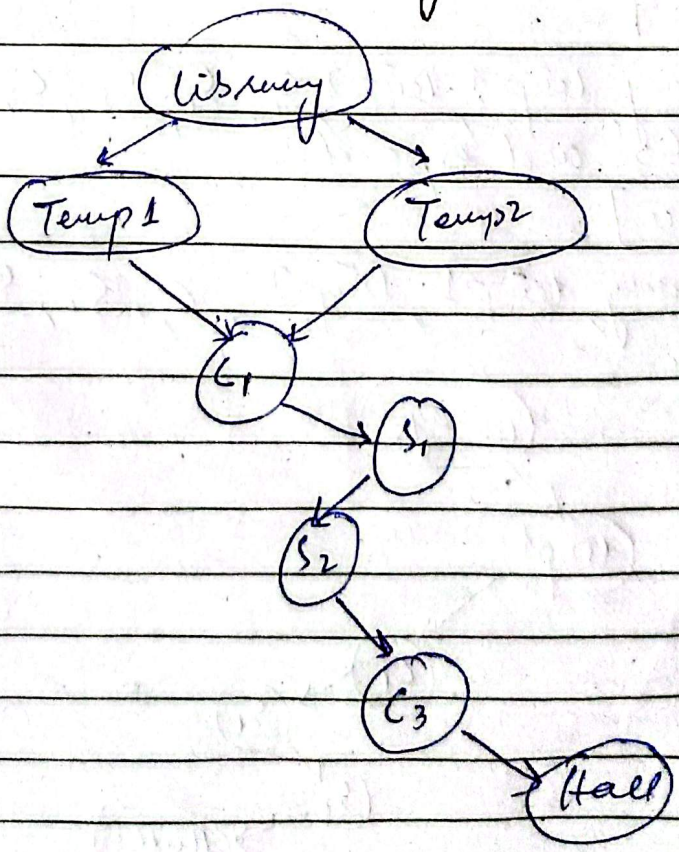
Sent No: BSCS, Sem 6th, Section B.
Artificial Intelligence

Problem: Go to Hall from Library.

Q2 Formulate the problem.

A (i) Initial State: Library
(ii) Goal State: Hall
(iii) State space: CS department
(iv) Action: Move from one state to another
(v) transition Model: To reach from library to Hall.
(vi) Cost. path: Distance between nodes d=1

Q3 Find path From library to hall.



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Date:

Q4 Find the shortest path.

A) Initialization:

(i) Q: [Library]
V: [Library]

(ii) deque Library and expand
Q: [Temp 1, Temp 2]
V: [Library, Temp 1, Temp 2]

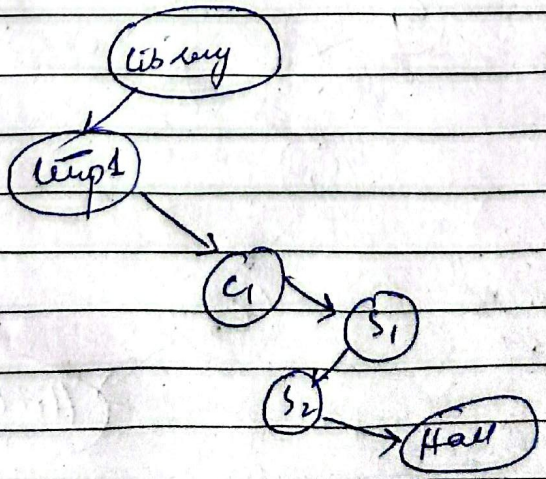
(iii) deque Temp 1 and expand
Q: [Temp 2, C1]
V: [Library, Temp 1, Temp 2, C1]

(iv) deque Temp 2 and C1 and expand
Q: [S1]
V: [Library, Temp 1, Temp 2, C1, S1]

(v) deque S1 and expand
Q: [S2]
V: [Library, Temp 1, Temp 2, C1, S1, S2]

(vi) deque S2 and expand
Q: [C3]
V: [Library, Temp 1, Temp 2, C1, S1, S2, C3]

(vii) deque C3 and expand
Q: [Hall]
V: [Library, Temp 1, Temp 2, C1, S1, S2, C3, Hall]



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CamScanner